Let $\{e_1, \dots, e_k\} \subseteq \mathbb{R}^n$ orthonormal, i.e. $\langle e_i, e_j \rangle = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{o.w.} \end{cases}$.

Let $v \in \mathbb{R}^n$, then. $\|v\|^2 \geq \sum_{i=1}^{k} |\langle v, e_i \rangle|^2$. (Equality holds if $v \in \text{Span}(\{e_1, \dots, e_k\})$

MA1522 Linear Algebra for Computing

Lesson 10: Basis and Dimension

Idea: You can extend $\{e_1, \dots, e_k\}$ to $\{e_1, \dots, e_k\} \cup \{e_{k+1}, \dots, e_n\}$ orthonormal basis of $\mathbb{R}^n$.

$\langle v, e_j \rangle = \langle \sum_{i=1}^{n} c_i e_i, e_j \rangle$ then c_j.

$\|v\|^2 = \sum_{i=1}^{n} c_i^2$.

$\sum_{i=1}^{k} |\langle v, e_i \rangle|^2 = \sum_{i=1}^{k} c_i^2$

$v = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$. $\|v\|^2 = \sum_{i=1}^{k} a_i^2$.

$\sum_{i=1}^{k} |\langle v, e_i \rangle|^2 \sim \sum_{i=1}^{k} a_i^2$.

Announcement: Online Quiz 2

- **Quiz 2 covers Chapter 2**, Sections 2.1–2.10: matrix algebra, matrix equations, inverses and elementary matrices, LU factorization, and determinants.

- It opens tomorrow, **Saturday 12 September, 0:00**, and is due **Sunday 13 September, 23:59**.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson10` (e.g. `A1234567Z_lesson10.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 10**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



Section 3.6 Basis

Definition

Let V be a subspace of $\mathbb{R}^n$. A set $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\} \subseteq V$ is a **basis** for V if

(i) S spans V, that is $\text{span}(S) = V$, and

(ii) S is linearly independent.

$\text{Span}(S) = \text{Span}\{u_1, \ldots, u_k\}$

Theorem

Suppose S is a basis for V. Then every vector $\mathbf{v} \in V$ can be written as a linear combination of vectors in S uniquely.

(i) : $V = \text{Span}(S)$: $V = c_1 u_1 + c_2 u_2 + \cdots + c_k u_k$ $\quad c_i \in \mathbb{R}$.

(ii) S is linearly independent $\Longrightarrow$ $c_1, \ldots, c_k$ is unique.

$v \in V \longleftrightarrow [v]_S = \begin{pmatrix} c_1 \\ \vdots \\ c_k \end{pmatrix}$: coordinate of v w.r.t. S.

Section 3.7 Dimension

Thm: If S_1, S_2 both basis of $V \subseteq \mathbb{R}^n$, then the size of S_1 = the size of S_2.

Definition

Let V be a subspace of $\mathbb{R}^n$. The **dimension** of V, written $\dim(V)$, is the number of vectors in any basis of V.

Theorem

Let A be an $m \times n$ matrix. The number of non-pivot columns in the reduced row-echelon form of A is the dimension of the solution space $V = \{ \mathbf{u} \in \mathbb{R}^n \mid A\mathbf{u} = \mathbf{0} \}$.

If $S \subseteq V$ and $|S| > \dim(V)$, then S is linearly dependent; if $|S| < \dim(V)$, then S cannot span V.

Question 1

Let V be the solution space of the homogeneous linear system

$$\begin{cases} x_1 + x_2 + 3x_3 + 2x_4 = 0 \\ x_1 - x_2 + x_3 - 4x_4 = 0 \\ 2x_1 + 3x_2 + 7x_3 + 7x_4 = 0 \end{cases}$$

Find a basis S for V, and find $\dim(V)$.

Question 1 — solution

$$\left(\begin{array}{cccc|c} 1 & 1 & 3 & 2 & 0 \\ 1 & -1 & 1 & -4 & 0 \\ 2 & 3 & 7 & 7 & 0 \end{array}\right) \xrightarrow{\text{rref.}} \left(\begin{array}{cccc|c} 1 & 0 & 2 & -1 & 0 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

x_3 x_4.

this step should be immediate for you.

general sol. use non-pivot column as parameters. $x_3 = t$, $x_4 = s$.

$$\begin{cases} x_1 + 2t - s = 0 \\ x_2 + t + 3s = 0 \end{cases} \Longrightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -2t + s \\ -t - 3s \\ t \\ s \end{pmatrix}. \quad t, s \in \mathbb{R}.$$

solution space.

$$\begin{pmatrix} -2t + s \\ -t - 3s \\ t \\ 0t + s \end{pmatrix} = t \begin{pmatrix} -2 \\ -1 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} 1 \\ -3 \\ 0 \\ 1 \end{pmatrix}. \Rightarrow \text{sol. space} = \text{Span}\left(\left\{\begin{pmatrix} -2 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -3 \\ 0 \\ 1 \end{pmatrix}\right\}\right)$$

S = S

Then one can check S is linearly independent.

Conclusion: S is a basis of the sol. space and $\dim(\text{sol. space}) = 2$.

Section 3.6 Basis for $\mathbb{R}^n$

Corollary

Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be a subset of $\mathbb{R}^n$ and let $A = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_k \end{pmatrix}$. Then S is a basis for $\mathbb{R}^n$ if and only if $k = n$ and A is invertible.

$n \times k$

So for a square matrix A of order n, the list of equivalent statements from Chapter 2 gains two more entries:

(x) the columns of A are linearly independent; $\Leftrightarrow$ $Ax = 0$ has only the trivial sol.

(xi) the columns of A span $\mathbb{R}^n$. $\Leftrightarrow$ $Ax = v$ is always consistent $\forall v \in \mathbb{R}^n$.

Question 2

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ and $T = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4\}$ be sets of vectors in $\mathbb{R}^4$, and let $A = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 & \mathbf{w}_1 & \mathbf{w}_2 & \mathbf{w}_3 & \mathbf{w}_4 \end{pmatrix}$. Suppose that

$$\operatorname{rref}(A) = \begin{pmatrix} 1 & 0 & 0 & 2 & 0 & 1 & 3 \\ 0 & 1 & 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 2 \end{pmatrix}.$$

Hint: Use elementary matrix and block mult.

Which of the following are true? Justify each answer.

(a) S is linearly independent. $\iff$ rref $\begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{pmatrix}$ every column is pivot.

(b) T is a basis for $\mathbb{R}^4$. $\iff C = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 & \mathbf{w}_1 \end{pmatrix}$ is invertible.

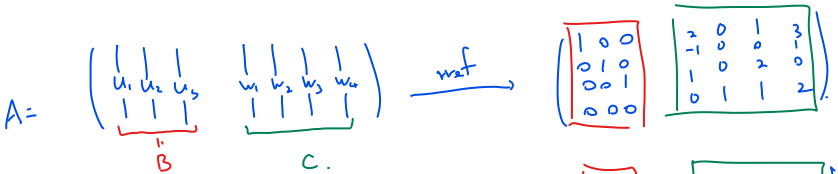
(c) $S \cup \{\mathbf{w}_1\}$ is a basis for $\mathbb{R}^4$.

(d) $S \cup \{\mathbf{w}_2\}$ is a basis for $\mathbb{R}^4$.

a) Question: How to find rref $\begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{pmatrix}$ from rref(A)?

b) Check invertibility of B from rref(A).

Question 2 — solution: (a) and (b)



Namely, $E_k E_{k-1} \cdots E_1 \left(\begin{array}{c|c} B & C \end{array}\right) = \left(\begin{array}{c|c} \vdots & \vdots \end{array}\right)$.

$E = E_k E_{k-1} \cdots E_1$ — 4×4 invertible. A.

Block mult gives

$E \cdot B = \begin{pmatrix} 1&0&0 \\ 0&1&0 \\ 0&0&1 \\ 0&0&0 \end{pmatrix} \Longrightarrow \operatorname{rref}(B) = \begin{pmatrix} 1&0&0 \\ 0&1&0 \\ 0&0&1 \\ 0&0&0 \end{pmatrix}$

S is linearly independent since every column in $\operatorname{rref}(B)$ is pivot.

$E \cdot C = \begin{pmatrix} 2&0&1&3 \\ -1&0&0&1 \\ 1&0&2&0 \\ 0&1&1&2 \end{pmatrix} \Longrightarrow \operatorname{rref}(C) = \operatorname{rref}(E \cdot C) \overset{\text{Matlab}}{=\!=} \begin{pmatrix} 1&0&0&0 \\ 0&1&0&0 \\ 0&0&1&0 \\ 0&0&0&1 \end{pmatrix}$.

$\Longrightarrow T$ is linearly independent.

$\Longrightarrow T$ is a basis for $\mathbb{R}^4$.

Question 2 — solution: (c) and (d)

$$\text{rref}\begin{pmatrix} | & | & | & | & | & | & | \\ u_1 & u_2 & u_3 & w_1 & w_2 & w_3 & w_4 \\ | & | & | & | & | & | & | \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 2 & 0 & 1 & 3 \\ 0 & 1 & 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 2 \end{pmatrix}.$$

D

$\text{rref}(D)$

$$=\begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

c) $S \cup \{w_1\}$.

From $\text{rref}(A)$, we see that D is row equivalent to the matrix above.

$\Rightarrow S \cup \{w_1\}$ is NOT linearly independent since $\text{rref}(D)$ has non-pivot column.

d) $S \cup \{w_2\}$. ... look at $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$. $\Rightarrow S \cup \{w_2\}$ forms a basis.

Question 3

Let V be the following subset of $\mathbb{R}^4$:

$$V = \left\{ \begin{pmatrix} a+d \\ a+b+2d+e \\ b+c+d+2e \\ c+e \end{pmatrix} \;\middle|\; a,b,c,d,e \in \mathbb{R} \right\}.$$

subspace $\Leftrightarrow V = \mathrm{span}(S)$ for some $S \subseteq \mathbb{R}^4$.

1. Show that V is a ~~vector space~~, and find a basis for V and $\dim(V)$.

2. Let $\mathbf{s}_1 = (1,1,1,1)^T$, $\mathbf{s}_2 = (1,2,2,1)^T$ and $\mathbf{s}_3 = (0,1,0,-1)^T$. Show that $S = \{\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3\}$ is also a basis for V.

3. Find a set T of vectors in $\mathbb{R}^4$ such that $S \cup T$ is a basis for $\mathbb{R}^4$.

size 3 size 1.

Question 3 — solution: part 1

$$V = \left\{ \begin{pmatrix} a & & d \\ a+b & & 2d+e \\ & b+c & d+2e \\ & c & e \end{pmatrix} \;\middle|\; a,b,c,d,e \in \mathbb{R} \right\} = \operatorname{span}\left(\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \\ 1 \end{pmatrix} \right\} \right)$$

$\Rightarrow V$ is a subspace.

$$a\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + b\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c\begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} + d\begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} + e\begin{pmatrix} 0 \\ 1 \\ 2 \\ 1 \end{pmatrix}.$$

$$\begin{array}{cccccc} u_1 & u_2 & u_3 & u_4 & u_5 \\ \| & \| & \| & \| & \| \\ \end{array}$$

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix} \xrightarrow{\text{ref.}} \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

pivot.

$\{u_1, u_2, u_3\}$ is linearly independent.

and $\operatorname{span}(\{u_1, u_2, u_3\}) = \operatorname{span}(\{u_1, \ldots, u_5\}) = V$.

$\Rightarrow \{u_1, u_2, u_3\}$ forms a basis of V.

$\dim(V) = 3$.

Question 3 — solution: part 2

$w_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$ $w_2 = \begin{pmatrix} 1 \\ 2 \\ 2 \\ 1 \end{pmatrix}$ $w_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}$.

Show that $\{w_1, w_2, w_3\}$ is also a basis of V.

(i) $\text{Span}(\{w_1, w_2, w_3\}) = V = \text{Span}(\{u_1, u_2, u_3\})$

(ii) $\{w_1, w_2, w_3\}$ is linearly independent.

(i): $\text{Span}(\{w_1, w_2, w_3\}) \geq \text{Span}(\{u_1, u_2, u_3\}) \iff u_1, u_2, u_3 \in \text{Span}(\{w_1, w_2, w_3\})$

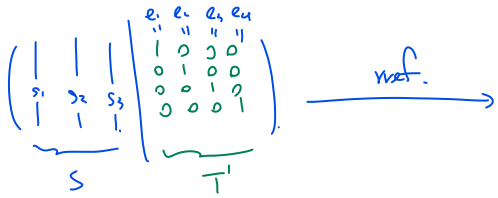
$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 1 & 1 \\ w_1 & w_2 & w_3 & u_1 & u_2 & u_3 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{array}\right) \xrightarrow{\text{rref}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & & & \\ 0 & 1 & 0 & & & \\ 0 & 0 & 1 & & & \\ 0 & 0 & 0 & & & \end{array}\right)$$

check these are not pivot column.

similarly for $\text{span}(\{w_1, w_2, w_3\}) \leq \text{span}(\{u_1, u_2, u_3\})$

(ii) $\text{rref}(\{w_1, w_2, w_3\}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$.

Question 3 — solution: part 3



We know that T' is a basis of $\mathbb{R}^4$.

By the argument in Question 2, we see that

$$\operatorname{rref}\begin{pmatrix} | & | & | & | \\ s_1 & s_2 & s_3 & e_1 \\ | & | & | & | \end{pmatrix} = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} | & | & | & | \\ s_1 & s_2 & s_3 & e_1 \\ | & | & | & | \end{pmatrix} \text{ is invertible} \Rightarrow \{s_1, s_2, s_3, e_1\} \text{ is a basis.}$$

In fact, $\{s_1, s_2, s_3\} \cup \{e_i\}$ with $i = 1, 2, 3, 4$ all form a basis.

Challenge

Let V be a subspace of $\mathbb{R}^n$, let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be a basis for V, and let $T = \{\mathbf{w}_1, \ldots, \mathbf{w}_k\} \subseteq V$ be a set of k vectors in V.

(a) Show that if T is linearly independent, then $\text{span}(T) = \text{span}(S) = V$.

(b) Show that if $\text{span}(T) = V$, then T is linearly independent.

In particular, T is linearly independent if and only if T spans V: for a set of exactly k vectors in V, either one on its own already makes T a basis.

Next lesson

- **Sections 3.6–3.8 Coordinates, Dimensions, Transition Matrices**

Please watch the video clips for these sections beforehand.